Module 03: Perception — Bayesian Mechanics and the Free Energy Principle

> **Course**: Active Inference 401 | **Unit**: Advanced Theory | **Audience**: Graduate students / researchers

Learning Objectives

1. Analyze **Bayesian mechanics** — the physics of systems that can be described as performing inference.
2. Evaluate the **particular partition** (internal, external, sensory, active states) and its consequences for autonomous systems.
3. Examine the mathematical conditions under which physical dynamics can be read as **belief updating**.

Key Concepts

1. The Particular Partition (Markov Blanket Formalism)

The foundation of the Free Energy Principle is a statistical partition of a system's states into four sets:

External states (η): States of the environment that are hidden from the agent — the true causes of observations.

Sensory states (s): States that are influenced by external states but not by internal states. These are the "input" channel — how the environment affects the agent.

Active states (a): States that are influenced by internal states but not by external states. These are the "output" channel — how the agent affects the environment.

Internal states (μ): States that are statistically separated from external states by the blanket (sensory + active) states. Internal states can only "see" external states through the blanket.

The Markov blanket condition: Internal states are conditionally independent of external states given the blanket states: $p(\mu \mid \eta, s, a) = p(\mu \mid s, a)$. This statistical separation is what creates the "inside" and "outside" of a system.

2. The Free Energy Lemma

Under the particular partition, if the system is at nonequilibrium steady state (NESS), then the internal states can be shown to parameterize a probability distribution over external states:

The remarkable result: The flow of internal states ($d\mu/dt$) can be expressed as a gradient descent on variational free energy:

$d\mu/dt \approx -\partial F/\partial \mu$

where F is the variational free energy of the distribution $q(\eta; \mu)$ that internal states parameterize, with respect to the observations provided by sensory states.

What this means: Any system with a Markov blanket whose dynamics reach a steady state can be described *as if* its internal states are performing approximate Bayesian inference about external states. The system doesn't need to "know" it's doing inference — its physical dynamics are mathematically equivalent to inference.

3. Nonequilibrium Steady States and Flows

The mathematical framework requires understanding nonequilibrium statistical mechanics:

Steady-state density: At NESS, the system has a time-invariant probability density $p(x)$ where $x = (\eta, s, a, \mu)$, but the dynamics are not at thermodynamic equilibrium — there are circulating (solenoidal) currents.

Helmholtz decomposition: The flow of any state can be decomposed into:

$dx/dt = (Q - \Gamma) \nabla \ln p(x) + w$

where Q is the solenoidal (conservative) component, Γ is the dissipative component, and w is random fluctuation. The dissipative component performs gradient descent on the log-density (free energy minimization); the solenoidal component preserves the steady-state distribution while allowing non-trivial dynamics.

Solenoidal flow matters: The solenoidal component (Q) is crucial — it allows the system to explore its state space in complex, non-gradient ways while maintaining the steady-state density. Without solenoidal flow, all dynamics would be simple gradient descent, which cannot capture oscillatory or more complex behavior.

4. Path Integral Formulation

The dynamics can also be expressed in terms of paths (trajectories) over time:

Action and most likely path: The most likely path of the system minimizes an action functional:

$S[x(t)] = \int L(x, dx/dt)\, dt$

where the Lagrangian L is related to the free energy. This connects Active Inference to the principle of least action in physics.

Path integral free energy: The free energy can be expressed as a functional of entire trajectories, not just instantaneous states. This generalizes the static free energy to encompass dynamics:

$F[q(x(t))] = E_q[L] + H[q]$

This path-integral formulation is essential for understanding systems with memory and temporal dynamics.

5. The Bayesian Mechanics Research Programme

Bayesian mechanics extends the FEP into a formal physics of belief:

Claim: There exists a class of mechanical theories (Bayesian mechanics) that describe the physical dynamics of systems possessing Markov blankets in terms of the beliefs those systems hold about their environment.

Dual aspect monism: The same physical dynamics can be described in two equivalent ways: (1) in terms of physical forces, potentials, and dissipation (mechanics), or (2) in terms of beliefs, prediction errors, and free energy (inference). Neither description is more fundamental — they are dual aspects of the same process.

Gauge symmetry: The mapping from internal states to beliefs is not unique — there is a gauge freedom in how internal states parameterize distributions. This gauge freedom is analogous to gauge symmetry in physics and reflects the fact that different neural configurations can encode the same belief.

Summary

Bayesian mechanics formalizes the conditions under which physical systems can be described as performing inference. The particular partition (Markov blanket) creates an inside-outside boundary. The Helmholtz decomposition reveals gradient (inferential) and solenoidal (exploratory) flows. Path-integral formulations connect Active Inference to physics. The dual-aspect nature of Bayesian mechanics connects mechanics and inference.

Further Reading

- Friston, K. J. (2019). A free energy principle for a particular physics. *arXiv preprint* arXiv:1906.10184.
- Da Costa, L. et al. (2021). Bayesian mechanics for stationary processes. *Proceedings of the Royal Society A*, 477(2256), 20210518.
- Sakthivadivel, D. A. R. (2022). Towards a geometry and analysis for Bayesian mechanics. *arXiv preprint* arXiv:2204.11900.
- Parr, T. et al. (2022). *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press.